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SHORT COMMUNICATION

Social Context and Reaction to Separation in Peer-reared Pigtail Macaques: Some Preliminary Observations

MARIA L. BOCCIA, MARTIN L. REITE, KRISTINE KAEMINGK, POLLY HELD, and MARK L. LAUDENSLAGER
University of Colorado Health Sciences Center

ABSTRACT. The social environment affects both behavioral and physiological responses to separation from the mother. Less information is available on the impact of the social environment on the response to separation in peer-reared infant monkeys. This study reports the responses of peer-reared pigtail macaque infants to repeated separations, and the impact of social versus isolation housing during the separation. The responses of two pairs of monkeys were studied during four three-day separations. One of each pair was housed in isolation during the separation, and the other was with another pair of peers, with whom they had been living for one month prior to the separation. The isolation-housed peer responded to the separation with behavioral agitation, but no depression. The socially-housed peer's behavior did not differ from baseline during the separation. During successive reunions, all the separated monkeys, regardless of housing condition, exhibited declining levels of behaviors related to maintaining proximity to their attachment figure. Although the number of subjects is small, the results suggest that the presence of social support, in the form of a familiar peer, can ameliorate the response to separation, and that with repeated separations the responses of the monkeys changes significantly.

Key Words: Monkey; Primate; Depression; Attachment; Social support.

INTRODUCTION

Pigtail monkey infants who are separated from their mothers while living in social groups exhibit a behavioral agitation-depression reaction (KAUFMAN & ROSENBLUM, 1969) as well as disturbances in the regulation of autonom (heart rate and body temperature) activity and nocturnal sleep patterns (REITE, SHORT, SEILER, & PAULEY, 1981; REITE & SHORT, 1983). The presence of a peer in the social group during the mother's absence, however, appears to mitigate the separated infant's behavioral and physiological responses to separation (CAINE & REITE, 1981; HINDE & DAVIES, 1972; SUOMI et al., 1973), implicating the social environment as a factor in loss-related behavioral and physiological changes.

A number of studies have used peer separation to model depression, citing potential advantages such as improved accessibility of the subject, ability to perform repeated separations, and absence of a confounding developmentally changing mother-infant relationship (e.g. SUOMI et al., 1970; LEWIS et al., 1976; MINEKA & SUOMI, 1978). Results from these studies, however, are contradictory: Some researchers report the two stage response in peer separation similar to that seen with mother-infant separation (e.g. MINEKA et al., 1981;

SUOMI et al., 1970), while others (e.g. ANDERSON, 1983; BOCCIA et al., 1989; ERWIN et al., 1971; MAXIM, 1980; MINEKA et al., 1985) report agitation but no depression in response to peer separation. Those studies which find the second, depression stage of separation in peers typically employ repeated separation paradigms. This raises the question of what is happening during the repeated separations to the peer attachment system.

Furthermore, little data are available on the impact of the social environment on peer-separation responses. In our laboratory, peer-reared pigtail infants who have been separated from their peer and housed alone show evidence of behavioral and physiological agitation, but not depressive responses to the separation experience (BOCCIA et al., 1989; REITE, HARBECK, & HOFFMAN, 1981). In infant squirrel monkeys, this behavioral agitation response to separation from their peer is eliminated if they remain in their home environment with another familiar peer (HENNESSEY et al., 1982), again implicating the social environment in the response to separation. The pilot study reported here examines peer separation in pigtail macaque infants during four three-day separations, during which (1) the impact of social-versus isolation-housing on the separation response; and (2) the peer attachment system during repeated separations were assessed.

METHODS

Subjects were four female pigtail macaques (*Macaca nemestrina*) who were removed from their mothers at a mean age of 6 ± 5 (S.D.) days, and were reared together in two separate pairs with a peer who differed in age by no more than 6 days. Beginning at approximately 6 months of age, the two pairs of peers were housed together in a larger cage. At a mean age of 33 ± 4 weeks, the infants were implanted with multichannel biotelemetry systems that transmitted multivariable physiological data from the unrestrained infants (REITE & SHORT, 1983). After two weeks of recovery from surgery, with a total of 35 days of group housing, and a week of baseline observations, one pair was subjected to four cycles of three days of separation followed by four days of reunion. One member was isolated from the group, and the other pair member remained with the other peer-pair, but was separated from the animal with which it had been reared. Two weeks later, the same procedure was followed for the second pair, so that two of the infants experienced separation in an isolated and two in a social context. The experimental protocol is outlined in Table 1. Behavioral (see Table 2), physiological [heart rate, body temperature, and sleep patterns, obtained via

Table 1. Experimental protocol.

	Animal I.D.				
	Pair one		Pair two[1)]		
Age of Pair one	17.3.4	34.1.6	63.9	8.10	Age of Pair two
0–6 days	..... Removed from mother				0–6 days
1–6 months	..... Housed with peer partner				1–5 months
6.5 months	..... Both pairs housed together as group of four				6 months
7 months	Implant	Implant	Control	Control	6.5 months
7.5 months	4 separations (housed with pair two)	4 separations (housed alone)	Control	Control	7 months
8.5 months	Explant	Explant	Implant	Implant	8 months
9.5 months	Control	Control	4 separations (housed alone)	4 separations (housed with pair one)	9 months

1) One-half month younger than pair one.

Table 2. Behaviors used in this study.

Behavior	Definition
Activity measures:	
Activity	Number of times infant crossed imaginary borders dividing pen into 6 equal cubes
Locomotion	Movement in any direction of at least 15 cm and involving limb movement
Environmental exploration:	
Object explore	Visual or manual inspection of environment
Self-directed behaviors:	
Self-orality	Self-mouthing or self-licking
Scratch	Common usage
Social behaviors:	
Coo	Long, tonal vocalization
Squeal	Gecker (high pitched distress call)
Touch	Physical contact with another animal
Proximity	Within 60 cm of another animal
Initiate play	Any play solicitation
Social inspect	Close visual, tactile, oral olfactory inspection of non-genital region of another animal
Approach	Enter proximity to another animal
Leave	Leave proximity to another animal

implantable biotelemetry (PAULEY & REITE, 1981)], and primary and secondary antibody responses to keyhole limpet hemocyanin (KLH), *in vitro* mitogen stimulation and natural cytotoxicity were examined for each phase of the experiment (see BOCCIA et al., 1989 for detailed procedures). Physiological data could only be obtained for three animals (no data were obtained from one of the isolation-housed monkeys, due to equipment failure).

Analyses of Variance (ANOVAs) with Social/Isolation Housing, Phase, and Separation (first through fourth) as factors were conducted. Phase of the experiment was subdivided into the following: separation day one (SD1), the remainder of separation (S), reunion day one (RD1), and the rest of reunion (R). Post-hoc comparisons employed Newman-Keuls tests.

RESULTS

Prior to separation (i.e. during the baseline) the infant monkeys still preferred the monkey with whom they were reared over the new peers introduced four weeks previously. This was evident by significantly greater levels of Play, Proximity, and Contact with this monkey over the other monkeys (see Table 3). This suggests that, despite the month of experience with the second pair of peers, the monkey with whom they were reared was still their primary attachment figure.

Table 3. Behavior directed by individuals to primary attachment figure over other monkeys during the baseline period.

Behavior[1)]	Primary attachment	Other monkeys[2)]	*F* ratio (1/3 *df*)	*p* value
Play	5.01	1.13	102.42	.0021
Proximity	17.78	5.27	423.27	.0003
Contact	14.83	4.61	1031.90	.0001

1) Frequency/1,000 sec; 2) scores divided by two, because there were two other monkeys.

Table 4. Behaviors directed by separated peers to social partners during the four separation periods.*

Behavior	Separation 1	Separation 2	Separation 3	Separation 4
Play	1.12	2.71	1.44	1.56
Proximity	11.92	14.63	10.52	9.98
Contact	10.20	13.31	9.89	9.75

*These data were collected and analyzed in the same way as those presented in Table 1.

The isolated infants, in contrast to infants socially housed during the separation, were more active during the separation than during baseline or reunion phases, and especially

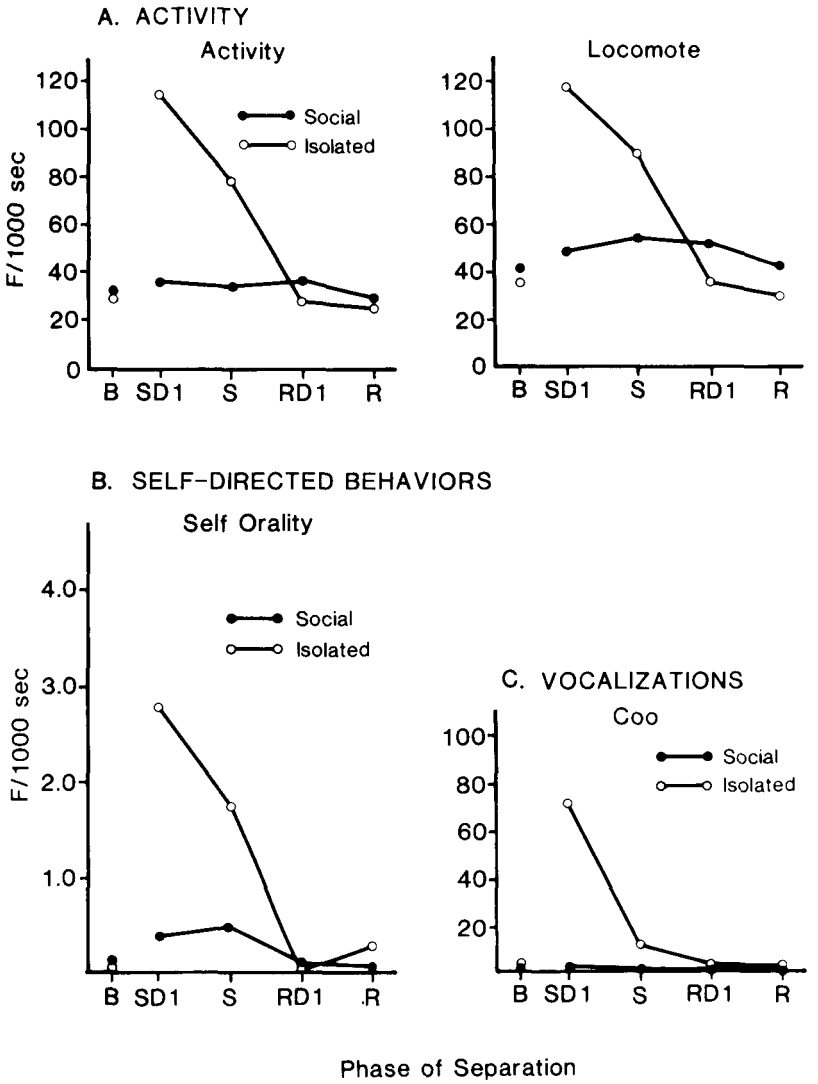


Fig. 1. Changes in A. activity measures, and B. self-directed behaviors, and C. coo vocalizations. B: baseline; SD1: first day of separation; S: remainder of separation period; RD1: first day of reunion; R: remainder of reunion.

active on SD1 [Phase × Housing Condition interactions for Activity ($F=68.51$, 3/6 *df*, $p=.0000$), and Locomotion ($F=16.30$, 3/6 *df*, $p=.0027$) (See Fig. 1A)]. These changes were the same for all separations. The socially-housed separated infants did not show these changes in behavior. They interacted with the other peer pair, as indicated by the behaviors play, proximity, and contact (see Table 4). Proximity to the other subjects in particular increased dramatically during the separation.

Self-directed behaviors increased during separation. Self-orality was higher during the separations in the isolated peer but not in the socially housed peer ($F=10.52$, 3/6 *df*, $p=.0084$) (see Fig. 1B).

Coo vocalizations showed dramatic increases on SD1 and then declined for isolated but not socially housed separated peers ($F=20.40$, 3/6 *df*, $p=.0015$) (see Fig. 1C). This pattern was observed consistently over all the separations.

A number of behaviors related to maintaining the social relationship with the attachment figure showed decreasing frequencies over repeated reunions (see Fig. 2). These effects did not differ by housing condition during separation. Proximity ($F=12.09$, 3/6 *df*, $p=.0059$) showed a decline which was significant by the last reunion, while Touch ($F=4.14$, 3/6 *df*, $p=.0656$) showed a decline by the third reunion. Play Initiation ($F=6.89$, 3/6 *df*, $p=.0227$) and Social Inspect ($F=14.83$, 3/6 *df*, $p=.0035$) also showed declining levels over repeated reunions. Both Approach ($F=12.74$, 3/6 *df*, $p=.0052$) and Leave ($F=4.14$, 3/6 *df*, $p=.06$) declined with repeated reunions to below baseline levels by the fourth reunion. These measure active regulation of change in proximity to the attachment object, so the changes suggest a decrease in activity regarding the maintenance of proximity. Finally, these effects were probably not developmentally related declines in that the unseparated peer pair of similar age did not show this decline over the same time periods.

Social housing during separation also eliminated autonomic responses typical of those previously seen in the peer-separation paradigm, where both members of the peer pair were housed alone. There were no significant changes in HR, BT, or sleep in the socially housed animals. The isolated animal showed autonomic responses similar to those found previously for peers: (1) higher mean day HR (an increase of 55 bpm) and BT (an increase of .93°C) on the days of separations; (2) longer REM latencies (an increase of 145 min) on the night immediately after separation than on baseline or the other nights of separation. There were no effects of separation or housing on the primary or secondary antibody response to KLH or the *in vitro* immune responses measured.

DISCUSSION

Evidence of agitation but not of depression was found only in the isolation-housed separated monkeys. The socially-housed individuals continued to exhibit baseline levels of behavior and physiology, demonstrating no separation effect. In addition, all of the separated monkeys, whether socially- or isolation-housed during the separation, showed decreasing levels of proximity-maintaining behaviors with repeated reunions. These results bear on two questions: the importance of the social environment in modulating the response to separation, and the inconsistency in previous attempts to induce depressive behavior in peer-separation paradigms.

These results suggest that the context of separation, and specifically, the availability of social support (in the form of the presence of familiar peers), affects the infant's response to separation. The infants in the present study interacted with these familiar peers, by

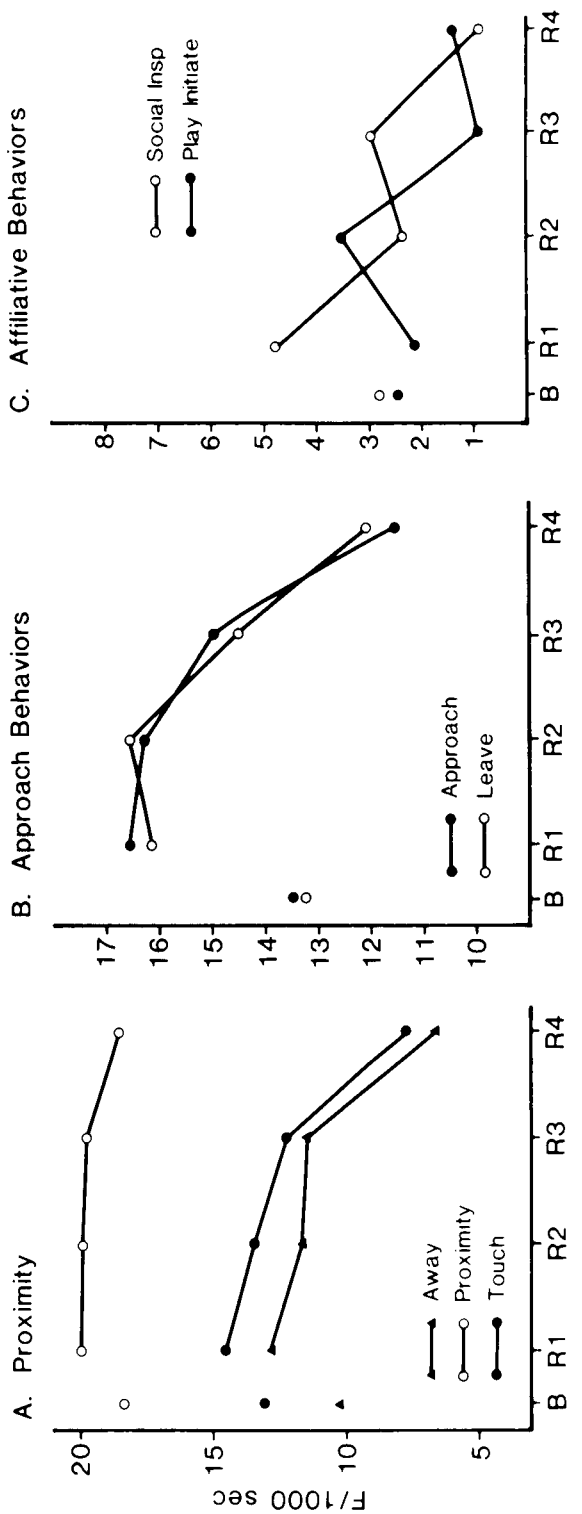


Fig. 2. Levels of A. proximity, B. approach, and C. affiliative behaviors during repeated reunion phases. (legend as in Fig. 1)

playing with and contacting them. Others have also shown that social housing of mother-separated infants attenuates the despair response (HINDE & DAVIES, 1972; SUOMI et al., 1973). The separated peers in the present study also showed an attenuated response (both behaviorally and physiologically) when they were housed with other monkeys. In fact, in this study, the socially-housed monkeys did not even evidence agitation in response to separation from their primary attachment figure. Other research also suggests that when social support from an alternative attachment is available, the response to separation is attenuated (e. g. REITE et al., 1989), but when the attachment is not present, the presence of other animals does not attenuate the separation response in monkeys (REITE, 1985; REITE et al., 1978). Thus, these subjects probably had developed some attachment to the other peer pair prior to the separation, so that their presence during separation mitigated the effect of the separation from their primary attachment figure.

The isolation housed animals evidenced agitation, but not depression. They showed elevated levels of coo vocalizations and self-orality, both of which have been interpreted to be attempts at behavioral coping, as evidenced by negative relationships between these behaviors and physiological indices of stress (e. g. BAYART et al., 1990; WIENER et al., 1990). Field studies of most Old World primates have suggested that coo vocalizations occur when an individual is separated from its group, and thus serves as a location call to re-establish contact. Hence its use by separated infants has been interpreted as an attempt to relocate the mother. Likewise, self-orality is a behavioral coping strategy in that it serves as a means of self-soothing under conditions of distress. The classic marker of behavioral depression, slouched posture, was never seen even in the isolation housed monkeys.

Not all studies using peer-separation paradigms observe depression. Despair responses have been found in peer-separation studies when the infants are separated repeatedly (up to 16 times: MINEKA et al., 1981; SUOMI et al., 1970). Our findings suggest a possible explanation of these differences. Over the course of the repeated reunions in this study, the proximity-maintaining behaviors declined, so that by the last separation in the series, these behaviors were far below baseline levels. Thus, with repeated separations the behavior of the monkeys toward their rearing partner changed. This change included a decrease in the behaviors regulating proximity. Repeated separations, then, may be producing a response similar to learned helplessness (MINEKA, 1982): the infants learn that they cannot control proximity to their attachment figure, resulting in decreased intensity of proximity behaviors. Eventually, as with a large number of separations noted above, this is manifest as a despair response. Alternatively, the attachment bond may be weakening, and the infants becoming more independent of one another. The question remains, however, what about the repeated separations experience is producing this change.

In contrast to other reports using separation paradigms (see LAUDENSLAGER, 1988), immunological changes were not observed in either *in vivo* or *in vitro* responses. It may be that the brief separation periods, without adequate time for recovery to baseline obscured possible immunomodulatory effects. Recent results from our laboratory (LAUDENSLAGER et al., 1990), suggest that immunological changes consequent to separation do not appear until the second week of separation. Hence these short (three day) separations may be of inadequate duration to find an immunological effect.

The number of subjects in this study is small and were all females, and the results, therefore, must be interpreted cautiously and may have limited generalizability. Nevertheless, they suggest that the presence of peers may ameliorate the effect of peer-separation, and that repeated separations may result in changes in the nature or expression of the attachment relationship. Additional studies will be needed to confirm these results.

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Authors' Names and Addresses: MARIA L. BOCCIA, MARTIN L. REITE, KRISTINE KAEMINGK, POLLY HELD, and MARK L. LAUDENSLAGER, *Department of Psychiatry C268–68*, University of Colorado, Health Sciences Center, 4200 East Ninth Avenue, Denver, Colorado 80262, U. S. A.